01 Split BF16

Separate exponent and side bytes.

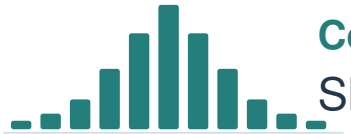
Exponent → encode

s | 8 exponent bits | 7 fraction bits

Sign + fraction → keep

02 Encode exponents

Assign bounded Huffman codes.



Common symbols
Short codes

Rare symbols
Escape + 8-bit literal

03 Store streams

Decode streams independently.

256 streams per block

Raw overflow tail

Sign + fraction bytes

04 Decode and use

Fused state updates

Weights + momentum → Decode + update → Encode new state

Exact reconstruction

$$\mathrm{decode}(\mathrm{encode}(u)) = u$$